PH 101

Lecture 11

OUTLINE

- More Worked out examples on Lagrange equations of motion
- Generalized Momentum

Double Pulley Problem

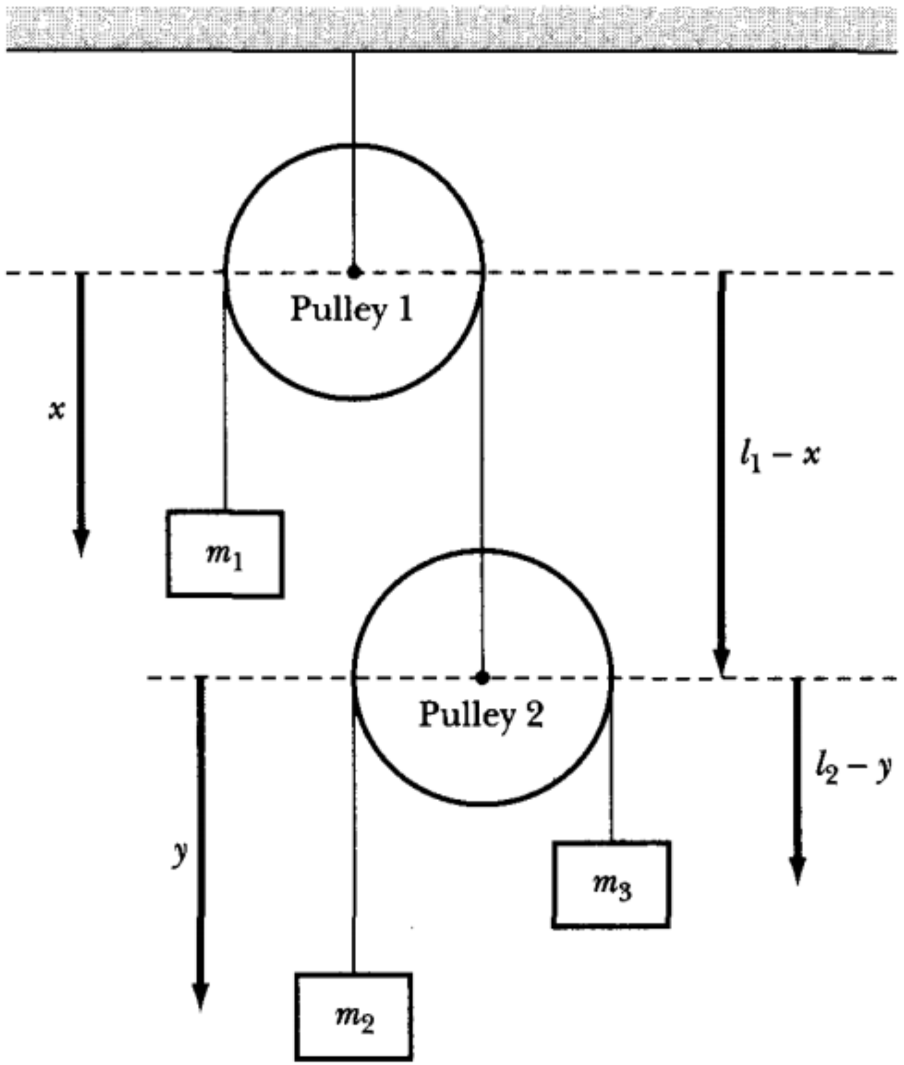
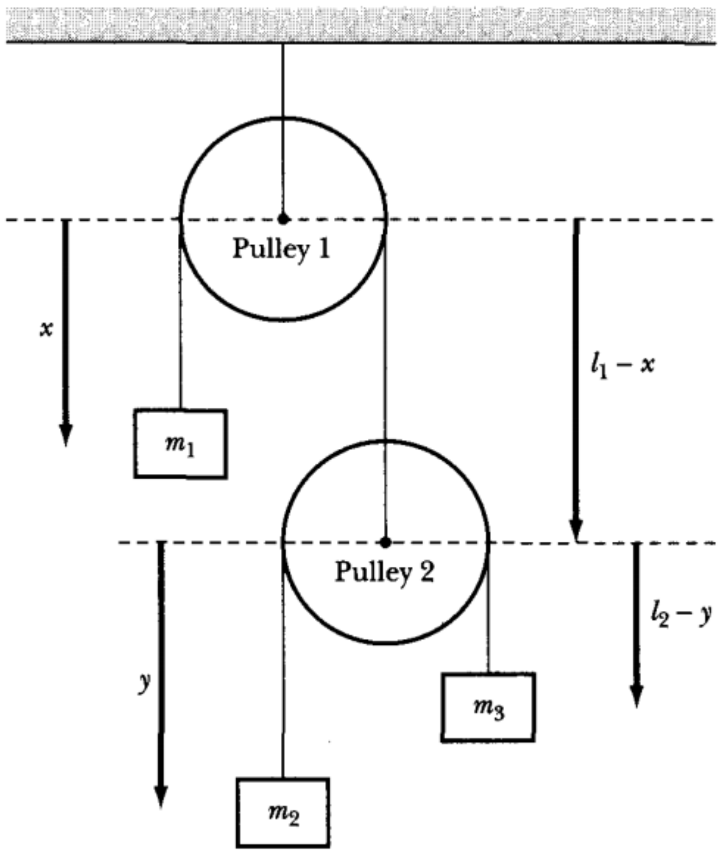


Fig. 1

Consider the double pulley system shown in Fig. 1. Use the coordinates indicted, and determine the equations of motion.

Solution. Consider the pulleys to be massless, and let l_1 and l_2 be the lengths of rope hanging freely from each of the two pulleys. The distances x and y are measured from the center of the two pulleys.



m_1:

$$v_1 = \dot{x}$$

m_2:

$$v_2 = \frac{d}{dt}(l_1 - x + y) = -\dot{x} + \dot{y}$$

m_3:

$$v_3 = \frac{d}{dt}(l_1 - x + l_2 - y) = -\dot{x} - \dot{y}$$

$$T = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 + \frac{1}{2}m_3 v_3^2$$

$$= \frac{1}{2}m_1 \dot{x}^2 + \frac{1}{2}m_2(\dot{y} - \dot{x})^2 + \frac{1}{2}m_3(-\dot{x} - \dot{y})^2$$

Let the potential energy $V = 0$ at $x = 0$

$$V = V_1 + V_2 + V_3$$

$$= -m_1 g x - m_2 g (l_1 - x + y) - m_3 g (l_1 - x + l_2 - y)$$

Thus,

$$L = T - V$$

$$= \frac{1}{2} m_1 \dot{x}^2 + \frac{1}{2} m_2 (\dot{y} - \dot{x})^2 + \frac{1}{2} m_3 (\dot{x} + \dot{y})^2 + (m_1 - m_2 - m_3) g x + (m_2 - m_3) g y + \text{constant}$$

Single particle in a central force field

The problem here is to use the Lagrange equations to generate the differential equations of motion for a particle constrained to move in a plane subject to a central force. The single constraint is given by $z = 0$, so we need two generalized coordinates. We choose plane polar coordinates: $q_1 = r$, $q_2 = \theta$. The transformation equations and resultant velocities are

$$x = r\cos\theta \qquad y = r\sin\theta$$

$$\dot{x} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta \qquad \dot{y} = \dot{r}\sin\theta + r\dot{\theta}\cos\theta$$

Thus, the Lagrangian is

$$T = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \tfrac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) \qquad V = V(r)$$

$$L = \tfrac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)$$

The relevant partial derivatives needed to implement the Lagrangian equations are

$$\frac{\partial L}{\partial \dot{r}} = m\dot{r} \qquad \frac{\partial L}{\partial r} = mr\dot{\theta}^2 - \frac{\partial V}{\partial r} = mr\dot{\theta}^2 + f(r)$$

$$\frac{\partial L}{\partial \theta} = 0 \qquad \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}$$

The equations of motion are, thus,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{r}} = \frac{\partial L}{\partial r} \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = \frac{\partial L}{\partial \theta} = 0$$

$$m\ddot{r} = mr\dot{\theta}^2 + f(r) \qquad \frac{d}{dt}(mr^2\dot{\theta}) = 0$$

A Particle Confined to Move on a Cylinder

Consider a particle of mass m constrained to move on a frictionless cylinder of radius R, given by the equation $\rho = R$ in cylindrical polar coordinates (ρ, θ, z) as shown in Figure 2. Besides the force of constraint (the normal force of the cylinder), the **only force** on the mass is a force $\vec{F} = k\vec{r}$ directed toward the origin. (This is the three dimensional version of the Hooke's-law force.) Using z and θ as generalized coordinates, find the Lagrangian L . Write down and solve Lagrange's equations and describe the motion.

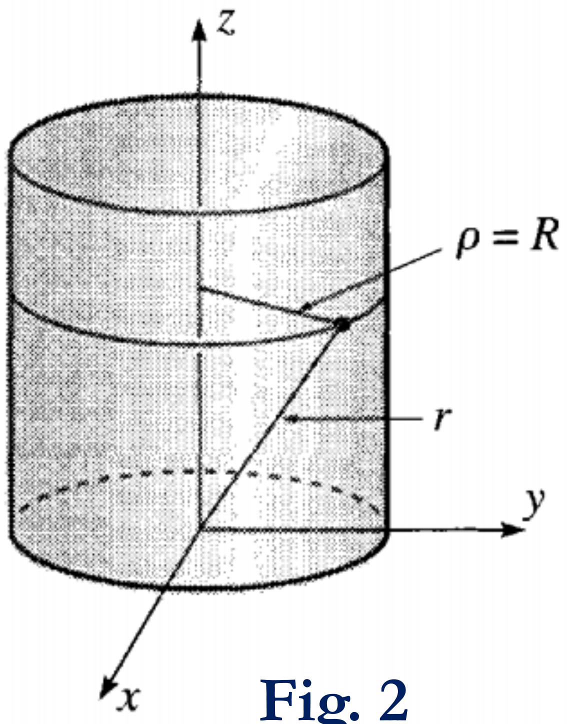
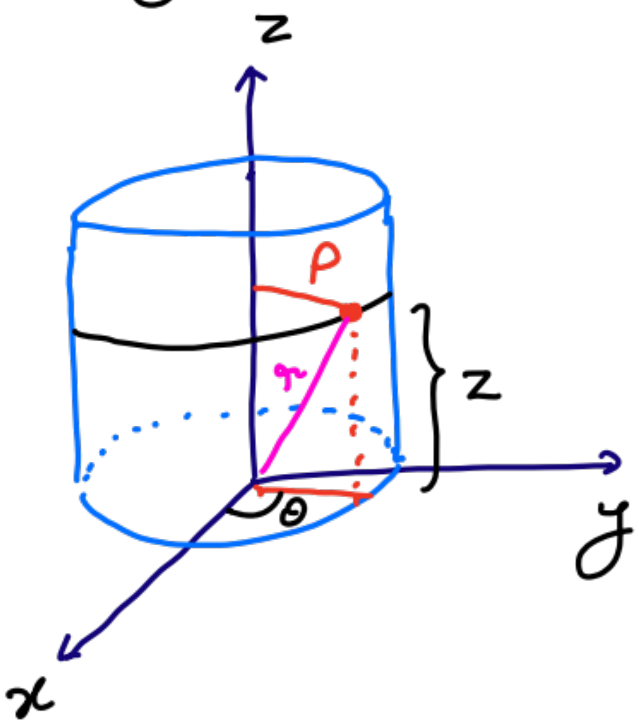


Fig. 2

<u>Please note</u>: Here we are using $(\boldsymbol{\rho}, \boldsymbol{\theta}, \boldsymbol{z})$ instead of $(\boldsymbol{r}, \boldsymbol{\theta}, \boldsymbol{z})$. Please see the diagram carefully. $\boldsymbol{\rho}$ is defined exactly the same as we defined $\boldsymbol{r}$ in cylindrical polar coordinate system. However, in this problem we define $\boldsymbol{r}$ as shown in the diagram.

Since the particle's coordinate is fixed at $\rho = R$, we can specify its position by giving just z and θ. The system has two degrees of freedom.



The velocity components are

$$v_\rho = 0 \quad ; \quad v_\theta = R\dot{\theta}$$

$$v_z = \dot{z}$$

Thus,

$$T = \frac{1}{2} m \left(R^2 \dot{\theta}^2 + \dot{z}^2 \right)$$

The potential energy for force $\vec{F} = -k\vec{r}$ is

$$V = \frac{1}{2}kr^2$$

(In analogy with SHM: $F = -kx$
$V = \frac{1}{2}kx^2$)

Now,

$$r^2 = \rho^2 + z^2$$
$$= R^2 + z^2$$

Thus,

$$L = \frac{1}{2}m\left(R^2\dot{\theta}^2 + \dot{z}^2\right) - \frac{1}{2}k\left(R^2 + z^2\right)$$

Since the system has two degrees of freedom, there are two equations of motion.

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{z}}\right)-\frac{\partial L}{\partial z}=0 \implies m\ddot{z}=-Kz \rightarrow (i)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right)-\frac{\partial L}{\partial \theta}=0 \implies \frac{d}{dt}\left(mR^2\dot{\theta}\right)=0 \rightarrow (ii)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{z}}\right) - \frac{\partial L}{\partial z} = 0 \implies m\ddot{z} = -kz \rightarrow (i)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0 \implies \frac{d}{dt}\left(mR^2\dot{\theta}\right) = 0 \rightarrow (ii)$$

(i) tells us that the mass executes simple harmonic motion in z-direction

(ii) tells us that the quantity $mR^2\dot{\theta}$ is constant, that is the angular momentum about z-axis is conserved. Because ρ is fixed, this means that $\dot{\theta}$ = constant and the mass moves around the cylinder with constant angular velocity $\dot{\theta}$, at the same time that it moves up and down in the z-direction in simple harmonic motion.

A Block Sliding on a Wedge

Consider the block and wedge shown in Figure 3. The block (mass m) is free to slide on the wedge, and the wedge (mass M) can slide on the horizontal table, both with negligible friction. The block is released from the top of the wedge, with both initially at rest. If the wedge has angle α and the length of its sloping face is l, how long does the block take to reach the bottom?

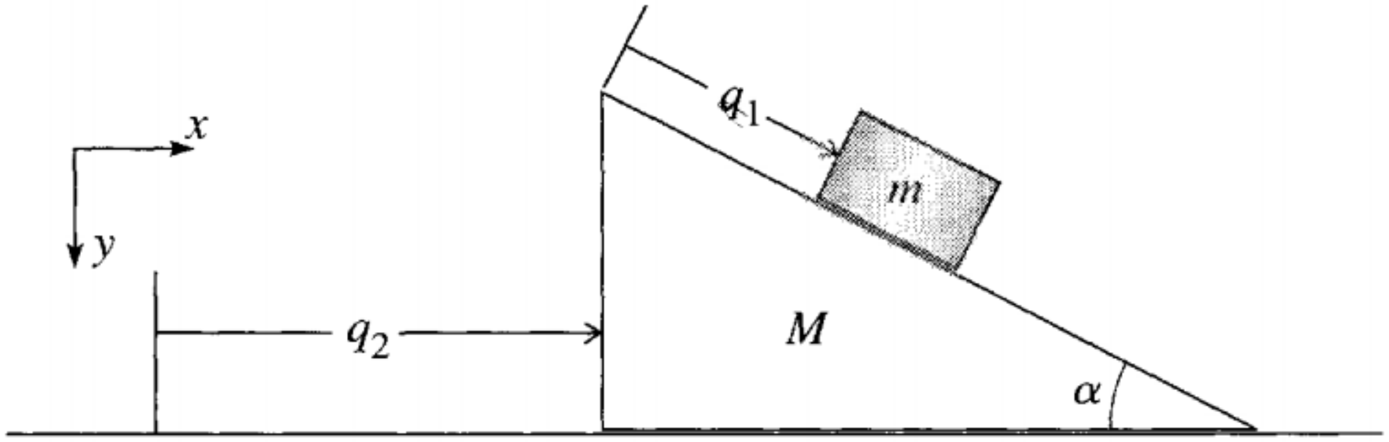
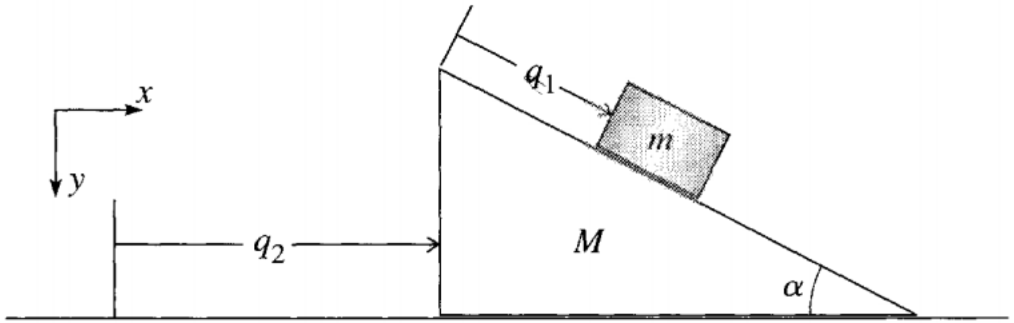


Fig. 3

A block of mass m slides down a wedge of mass M, which is free to slide over the horizontal table

Solution:

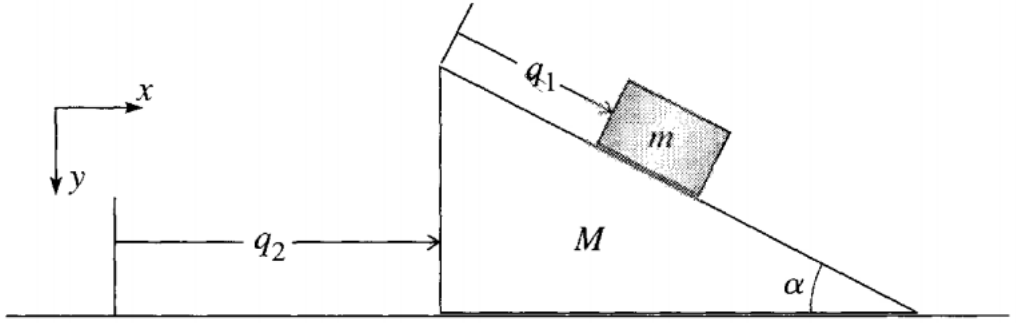
The system has two degrees of freedom, and a good choice of the two generalized coordinates is, as shown, the distance q_1 of the block from the top of the wedge and the distance q_2 of the wedge from any convenient fixed point on the table. The quantity we need to find is the acceleration $\ddot{q}_1$ of the block relative to the wedge, since with this we can quickly find the time required to slide the length of the wedge. Our first task is to write down the Lagrangian, and it is often safest to do this in Cartesian coordinates, and then rewrite it in terms of the chosen generalized coordinates.



The kinetic energy of the wedge is just

$$T_M = \frac{1}{2} M \dot{q}_2^2$$

The block's velocity relative to the wedge is $\dot{q}_1$ down the slope, but the wedge itself has a horizontal velocity $\dot{q}_2$ relative to the table. The velocity of the block relative to the inertial frame of the table is the vector sum of these two. Resolving into rectangular components (x to the right, y downward), we find for the velocity of the block relative to the table:



$$\mathbf{v} = (v_x, v_y) = (\dot{q}_1 \cos\alpha + \dot{q}_2, \dot{q}_1 \sin\alpha)$$

Thus the kinetic energy of the block is

$$T_m = \tfrac{1}{2}m(v_x^2 + v_y^2) = \tfrac{1}{2}m(\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1\dot{q}_2\cos\alpha)$$

The total kinetic energy of the system is

$$T = T_M + T_m = \tfrac{1}{2}(M+m)\dot{q}_2^2 + \tfrac{1}{2}m(\dot{q}_1^2 + 2\dot{q}_1\dot{q}_2\cos\alpha)$$

The potential energy of the wedge is a constant, which we may as well take to be zero. That of the block is $-mgy$, where $y = q_1 \sin\alpha$ is the height of the block measured down from the top of the wedge. Therefore

$$V = -\mathrm{mg}\,\mathrm{q}_1 \sin\alpha$$

and the Lagrangian is

$$L = T - V = \tfrac{1}{2}(M+m)\dot{q}_2^2 + \tfrac{1}{2}m(\dot{q}_1^2 + 2\dot{q}_1\dot{q}_2\cos\alpha) + mgq_1\sin\alpha.$$

The Lagrange's equations of motion are:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_2}\right) = \frac{\partial L}{\partial q_2} \Rightarrow \boxed{M\dot{q}_2 + m\left(\dot{q}_2 + \dot{q}_1 \cos\alpha\right) = \text{const.}}$$

$\rightarrow$ (i)

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_1}\right) = \frac{\partial L}{\partial q_1} \Rightarrow \boxed{m\left(\ddot{q}_1 + \ddot{q}_2 \cos\alpha\right) = mg\sin\alpha}$$

$\rightarrow$ (ii)

From (i) and (ii) we can obtain:

$$\ddot{q}_1 = \frac{g\sin\alpha}{1 - \dfrac{m\cos^2\alpha}{M+m}} \qquad \rightarrow \text{(iii)}$$

The distance travelled down the slope in time t is $\frac{1}{2}\ddot{q}_1 t^2$.

The time taken to travel length ℓ is just $\sqrt{\frac{2\ell}{\ddot{q}_1}}$, with $\ddot{q}_1$ given by (iii)

Generalized Momenta

Consider a holonomic mechanical system with Lagrangian $L = L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t)$.
Then the quantity p_j defined by

$$p_j = \frac{\partial L}{\partial \dot{q}_j}$$

is called the ***generalised momentum*** *corresponding to the coordinate* q_j. *It is also called the momentum* ***conjugate*** *to* q_j.

Recall the situation where a particle of mass m is moving in a potential $V(r)$.

In Cartesian coordinates, the Lagrangian of the system can be written as:

$$L = \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right) - \mathrm{V}(r);$$

where $r = \sqrt{x^2 + y^2 + z^2}$

Generalized Momentum:

$$p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x}; \qquad p_y = \frac{\partial L}{\partial \dot{y}} = m\dot{y}; \qquad p_z = \frac{\partial L}{\partial \dot{z}} = m\dot{z}.$$

It is not necessary that the generalized force has the same dimension as that of the so-called 'momentum'.

Generalized Force

Consider a holonomic mechanical system with Lagrangian $L = L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t)$.
Then the quantity Q_j defined by

$$Q_j = \frac{\partial L}{\partial q_j}$$

is called the generalized force corresponding to the coordinate q_j.

You can relate it to more if you recall the situation where a particle of mass m is moving in a potential $V(r)$.

In Cartesian coordinates, we can express the Lagrangian of the system as:

$$L = \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right) - \mathrm{V}(r)$$

where $r = \sqrt{x^2 + y^2 + z^2}$

Generalized Force:

$$Q_x = \frac{\partial L}{\partial x} = -\frac{\partial V}{\partial x} = F_x; \quad Q_y = \frac{\partial L}{\partial y} = -\frac{\partial V}{\partial y} = F_y; \quad Q_z = \frac{\partial L}{\partial z} = -\frac{\partial V}{\partial z} = F_z.$$

Here the generalized forces are exactly same as the usual Force we encounter. However **it must be kept in mind**, it is not necessary that the generalized force has the same dimension as that of the so-called 'force'.

Examples

Given the Lagrangian of a system,

$$L = \tfrac{1}{2}M\dot{x}^2 + \tfrac{1}{2}m\left(\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\cos\alpha\right) + mgy\sin\alpha.$$

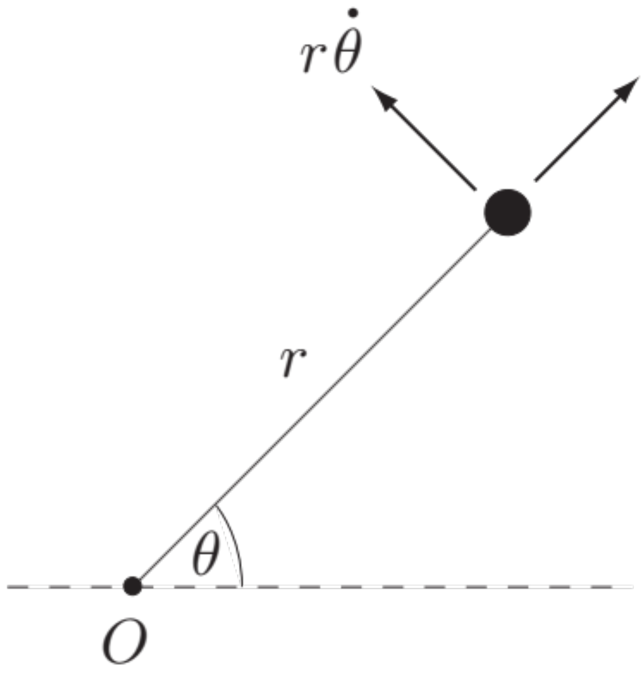
Find the generalised momenta.

Solution

With this Lagrangian, the momenta p_x and p_y are given by

$$p_x = \frac{\partial L}{\partial \dot{x}} = M\dot{x} + m\left(\dot{x} + \dot{y}\cos\alpha\right),$$
$$p_y = \frac{\partial L}{\partial \dot{y}} = m\left(\dot{y} + \dot{x}\cos\alpha\right). \blacksquare$$

A particle of mass **m** moves under the gravitational attraction of a fixed mass **M** situated at the origin. Take polar coordinates r, θ as generalized coordinates and obtain the Lagrangian of the system. Find the generalized momenta and force.



Solution:

The **kinetic energy** of the particle is

$$T = \tfrac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right)$$

and the **potential energy** of the particle (relative to infinity) is

$$V = -\frac{MG}{r}.$$

The **Lagrangian** of the particle is therefore

$$L = \tfrac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + \frac{MG}{r}.$$

Generalized Force:

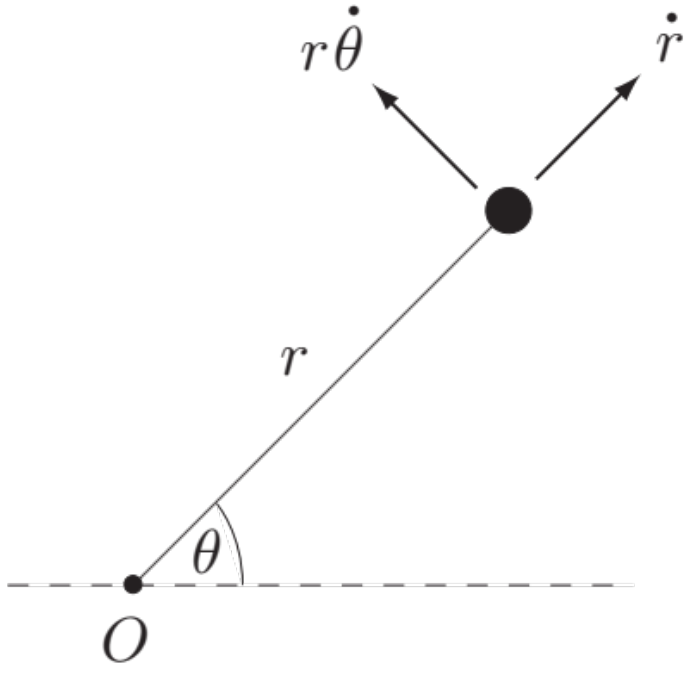
$$Q_r = \frac{\partial L}{\partial r} = mr\dot{\theta}^2 - \frac{MG}{r^2}$$

$$Q_\theta = \frac{\partial L}{\partial \theta} = 0$$

Generalized Momentum:

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}$$

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}$$



You can see that, here p_θ is the angular momentum of the particle about the axis through O perpendicular to the plane of motion.

p_r is the linear momentum along the radial direction.

In fact, by dint of Lagrange's equations:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = 0$$

We can see:

$$\frac{dp_j}{dt} = Q_j$$

"generalized force = rate of change of generalized momentum."